

TLA+ Study Group — Week 01

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1 Questions

1.1 *if* ($\Rightarrow$) and *iff* ($\Leftrightarrow$)

Definition

1. $\Leftrightarrow$ is the logical equivalence.
2. $p \Leftrightarrow q$ is read as p equivalent q or p if and only if q

Truth table

p	q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example

if ($\Rightarrow$) is one-way: If a number is divisible by 4, then it is even. (But not all even numbers are divisible by 4.)

iff ($\Leftrightarrow$) must hold both ways: A number is even $\Leftrightarrow$ it is divisible by 2. (Even $\Rightarrow$ divisible by 2, and divisible by 2 $\Rightarrow$ even.)

Note: An example of this in TLA+ can be found in `tla_plus/iff.tla`